TECHNICAL ARCHITECTURE SPECIFICATION • SYSTEM BLUEPRINT

AIOps Sentinel: Engineering Architecture & TRD

Multi-Tenant Telemetry Ingestion, Hybrid Inference, LangGraph Orchestration & Sandboxed Execution

1. High-Level Distributed Architecture

The architecture decouples telemetry producers from evaluation consumers. Target nodes forward raw metric tuples and standard error streams over TLS to a stateless FastAPI ingestion gateway. Data is buffered into Redis Streams for high-throughput backpressure dampening, evaluated through local ONNX-runtime ML models, routed into a LangGraph state machine for RAG-assisted diagnosis, and sandboxed inside Docker containers before remote actuation.

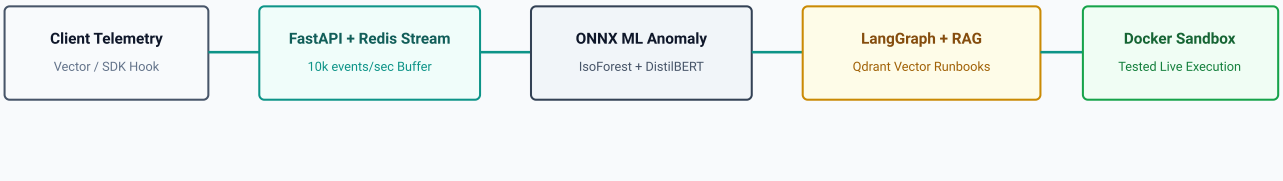


Figure 2.0: Core ingestion, inference, agentic orchestration, and sandboxed remediation topology.

2. Ingestion Pipeline & Telemetry Payload Contract

The ingestion service processes concurrent telemetry frames over HTTP/2. Incoming requests validate API authorization tokens against Redis key caches within 2 milliseconds. Telemetry batches are persisted into Redis Stream key `stream:telemetry:raw` with an event expiry TTL of 4 hours.

Field Name	Type	Validation & Schema Contract
`client_id`	UUIDv4 String	Unique tenant identifier mapped to authorization tokens and target policy configs.
`metrics.cpu_pct`	Float (0.0 - 100.0)	Kernel and user CPU utilization rolling average measured over 5-second intervals.
`metrics.mem_pct`	Float (0.0 - 100.0)	Resident set size (RSS) memory consumption against container hardware limits.
`log_stream`	Array of Objects	Array of log records: `{"ts": int, "lvl": "ERROR", "msg": str, "src": str}`.

3. Predictive Machine Learning Pipeline

Anomaly detection utilizes a two-tier hybrid model strategy:
Tier 1: Multivariate Metric Anomaly Engine. An unsupervised *Isolation Forest* model trained on standardized historical server metrics (`cpu_usage` , `memory_rss` , `io_wait` , `tx_error_rate`). Models are exported to ONNX format to maintain p99 evaluation latency under 5ms. Anomaly thresholds are normalized: values > 0.85 initiate active incident triage.
Tier 2: NLP Log Classifier. A quantized *DistilBERT* model fine-tuned on the LogHub benchmark classifies raw error strings into categorized operational states: `ROUTINE` , `DB_CONNECTION_EXHAUSTION` , `OOM_EARLY_WARNING` , and `FATAL_DEADLOCK` .

4. Generative AI & Retrieval-Augmented Generation (RCA)

When an anomaly score exceeds threshold bounds, the system queries Qdrant Vector DB containing chunked architecture blueprints, service runbooks, and past incident post-mortems using `bge-small-en-v1.5` 384-dimensional embeddings. The LLM processes the contextual prompt under a strict JSON schema contract specifying root cause justification, target system entities, and parameterized recovery shell scripts.

5. LangGraph State Machine Specification

Node Name	Input Dependencies	Execution Responsibilities	Next State
`diagnose_node`	Metrics + Raw Log Stream	Executes Qdrant vector retrieval and generates structured RCA payload.	`patch_gen`
`patch_gen_node`	RCA JSON + Runbook Rules	Drafts discrete shell commands matching system security whitelist.	`sandbox_test`
`sandbox_node`	Remediation Script	Executes dry-run in isolated Docker container. Inspects exit codes.	Condition Gate
`reflect_node`	STDERR Output + Exit Code	Reflects on execution errors and adjusts command flags (Max 3 iterations).	`patch_gen`
`dispatch_node`	Passed Sandbox Artifact	Sends remote Docker/K8s API commands or posts Slack approval webhook.	`END`

> **Self-Correction State Transition Rule**
> If `sandbox_exit_code != 0` and `retry_counter < 3` , transition directly to `reflect_node` with full error traces. If `retry_counter == 3` , terminate loop, abort auto-remediation, and trigger emergency Slack alert.

6. Sandbox Containerization & Safe Actuation

Remediation commands never execute directly against live production infrastructure without validation. The Sandbox Agent communicates with the local host daemon via the Docker SDK for Python. It initializes an ephemeral Alpine/Ubuntu container matching target OS dependencies with strict sandbox isolation flags: `--net=none` (complete network egress restriction), a non-root execution context, and a 15-second command timeout threshold.

7. PostgreSQL Relational State Schema

Table	Primary Fields	Constraints, Indices & Relational Foreign Keys
`clients`	id (UUID), name, api_key_hash	Primary key on `id`; Unique hash index on `api_key_hash`.
`incidents`	id (UUID), client_id, score, status	FK to `clients(id)` on delete cascade; Index on status and created_at.
`remediations`	id (UUID), incident_id, script, sandbox_pass	FK to `incidents(id)`; Boolean validation flag; retry count integer.
`audit_logs`	id (UUID), incident_id, stage, payload	JSONB payload preserving complete raw execution output and telemetry snapshots.

> **Security & AST Command Whitelist Guardrails**
> Prior to sandbox injection, generated scripts pass through an Abstract Syntax Tree (AST) parser. Commands containing destructive strings such as `rm -rf /`, `mkfs`, `dd`, `iptables -F`, or direct mutations to `/etc/passwd` raise an immediate security exception, quarantine the agent run, and flag the incident for human review.

8. Verification Matrix & Chaos Engineering

The verification suite executes automated end-to-end regression tests across three chaos scenarios:
• **Memory Leak Injection:** `stress-ng --vm 2 --vm-bytes 90%` to test proactive OOM container reboot and cache flush.
• **Connection Pool Exhaustion:** Continuous concurrent socket exhaustion against PostgreSQL port 5432 to verify dynamic connection limit expansion.
• **Cascading 5xx Storm:** Simulated HTTP gateway timeouts to validate replica autoscaling triggers.